SMART MEDICAL INSURANCE COST PREDICTOR WITH NGO & WELFARE AID GUIDANCE

Final Project Review | Department of Computer Science and Engineering | KLH CSE Bowrampet Campus

TEAM MEMBERS

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Guide: Ms. B. Pratyusha, Assistant Professor | Academic Year: 2026–27

Introduction & Problem Statement

Background

Healthcare costs in India keep rising and are hard to predict. Age, BMI and smoking change the cost a lot, so high-risk people find it hard to plan their money.

Problem Statement

Cost tools stop at an estimate. They do not tell a patient who cannot pay which NGO or welfare scheme can help, or by how much.

Motivation / Need

Low-income patients and families gain the most. Without a link between cost and aid, eligible people never find the help that exists.



Objectives & Scope

Objectives

1. Clean the data and run EDA to find the main cost drivers: age, BMI, smoking and city tier.
2. Encode features and compare Linear, Random Forest and Gradient Boosting models with MAE, RMSE and R^2.
3. Curate an NGO and welfare dataset and build a rule-based engine that matches users to eligible aid.
4. Build a dashboard that shows the cost estimate, the matched aid in ₹ and visual analytics.

Scope

In scope

- Cost prediction for Indian profiles, in ₹
- Rule-based matching to NGOs and welfare schemes
- Interactive dashboard with analytics

Out of scope

- Real insurance claims or policy pricing
- Live updates of scheme rules
- Medical advice or diagnosis

Literature Survey

S.No	Author(s) & Year	Method / Technique	Key Findings	Limitation / Gap
1	Orji et al., 2024	XGBoost, GBM, Random Forest with SHAP/ICE on 986 records	Tree ensembles predict well; smoking, BMI, age drive cost	Prediction only; no help for patients who cannot pay
2	IJRASET, 2023	Linear, Decision Tree, Gradient Boosting; Streamlit app	Boosting and tree models beat plain linear regression	One cost estimate; no aid guidance
3	Hong et al., 2025	Logistic Reg., Decision Tree, Random Forest on 1M+ ED bills	Decision Tree flagged 87% of patients needing aid	Finds need, but does not match NGOs or schemes
4	Rajpoot et al., 2025	AI recommendation system for government schemes	Helps citizens discover suitable schemes	No link to predicted medical cost



Research Gap

No existing work joins cost prediction with eligibility-matched NGO and welfare aid. Our project turns a predicted cost and a user profile into a ranked list of aid options (Objectives 3 and 4).

Proposed System & Methodology

STEP 1

Define Problem

Cost estimates alone do not help patients who cannot afford care.

STEP 2

Collect & Clean

Medical cost data (₹) and NGO/welfare data; clean, encode, scale.

STEP 3

EDA & Models

Study the data, then compare Linear, Random Forest and Gradient Boosting.

STEP 4

Match & Build

Rule-based aid matching and a Streamlit dashboard.

STEP 5

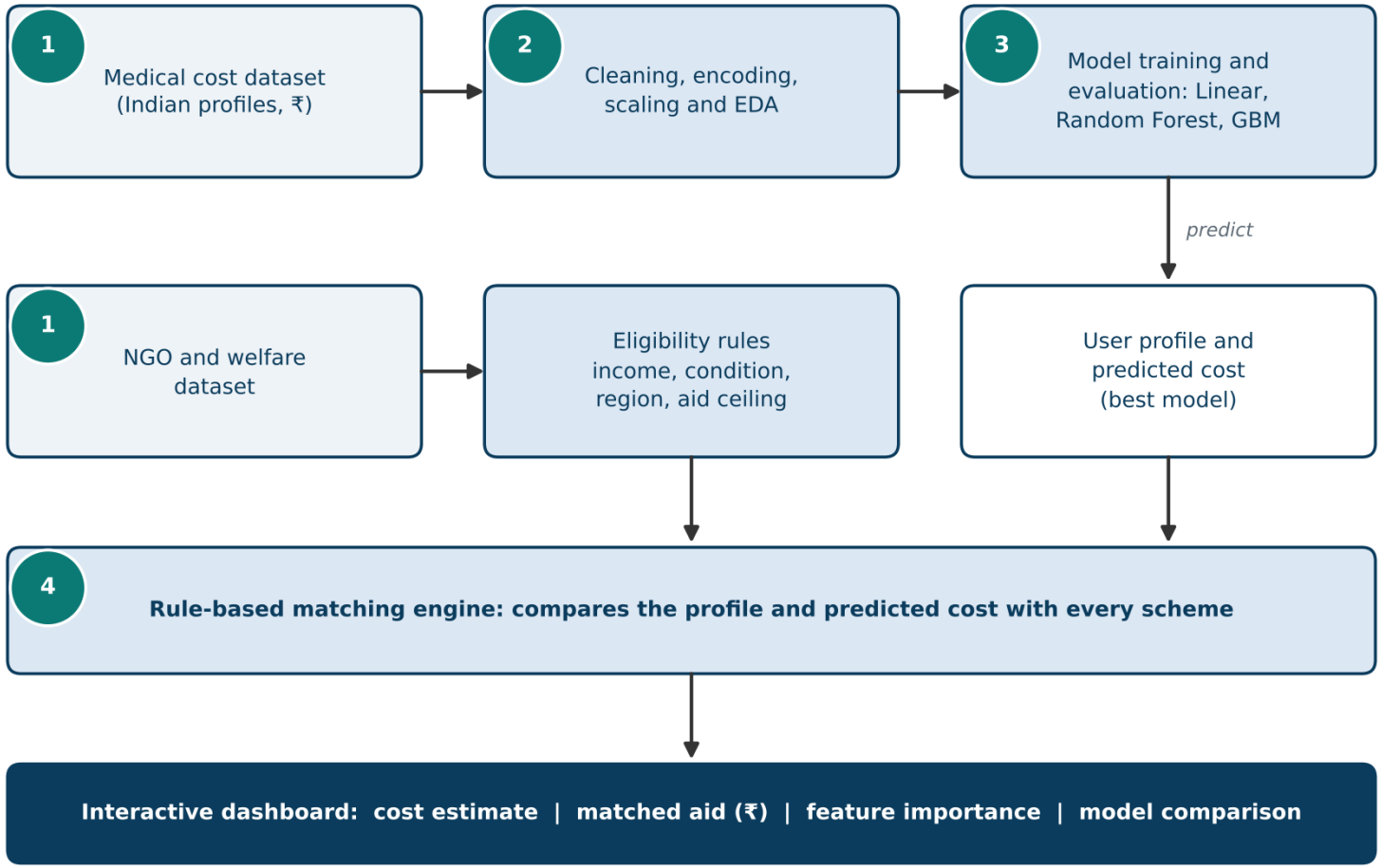
Test & Evaluate

MAE, RMSE, R^2, cross-validation and test cases.

Key Novelty / Contribution

Joins ML cost prediction with a transparent, rule-based match to NGO and government aid, using a self-curated eligibility dataset that no public source provides.

System Architecture



Components

1 **Data Layer**
Medical cost and NGO/welfare datasets

2 **Preprocessing & EDA**
Cleaning, encoding, scaling, plots

3 **Prediction Models**
Linear, Random Forest, Gradient Boosting

4 **Matching Engine & Dashboard**
Rules on income, condition, region; aid in ₹

Hardware & Software Requirements

Hardware

- Laptop or PC, Intel Core i5 or similar
- 8 GB RAM
- 256 GB SSD (datasets are small)
- No GPU needed (tabular data)

Software & Tools

- Windows 10/11
- Python 3.11
- Jupyter Notebook, VS Code, Google Colab
- Git and GitHub

Libraries / Frameworks

- pandas, NumPy
- scikit-learn
- matplotlib, seaborn
- Streamlit (dashboard)

Deployment / Platform: Local machine and Google Colab for training; Streamlit web app for the dashboard

Dataset / Data Understanding

2,500
Records (rows)

9
Features (columns)

75
Missing values (BMI)

1
Target (charges, ₹)

- Source: India-adapted medical cost data (self-prepared, amounts in ₹)
- Features: age, sex, BMI, children, smoker, region, city tier, income, pre-existing condition
- Quality: 75 missing BMI values (3%), right-skewed charges, a few high-cost outliers

india_medical_cost.ipynb – Jupyter

In [2]: **df.head(8)**

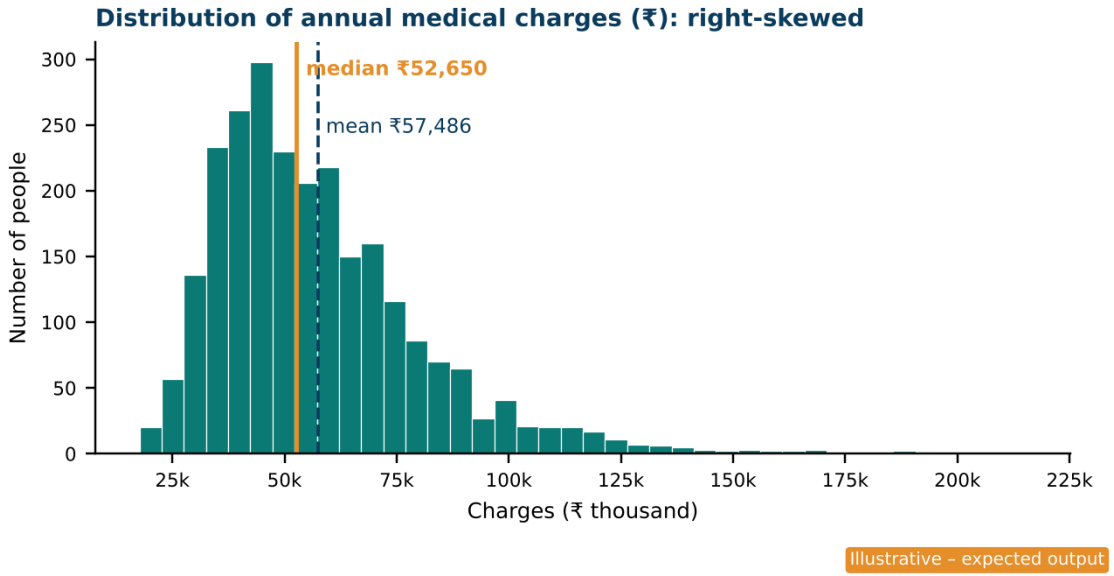
age	sex	bmi	children	smoker	region	city_tier	income(₹)	pre_existing	charges
42	Female	26.9	4	No	Central	Tier-2	111,000	No	59,600
26	Male	25.8	1	No	North	Tier-2	250,000	Yes	62,910
47	Male	28.8	0	No	East	Tier-2	786,000	Yes	94,610
49	Male	29.9	1	No	West	Tier-3	745,000	No	49,620
18	Male	29.8	0	No	North	Tier-3	621,000	No	33,340
22	Male	25.1	1	No	East	Metro	386,000	Yes	61,060
40	Female	32.0	0	No	West	Tier-3	486,000	No	50,820
34	Male	26.7	3	No	West	Tier-2	343,000	Yes	65,410

In [3]: **df.info()**

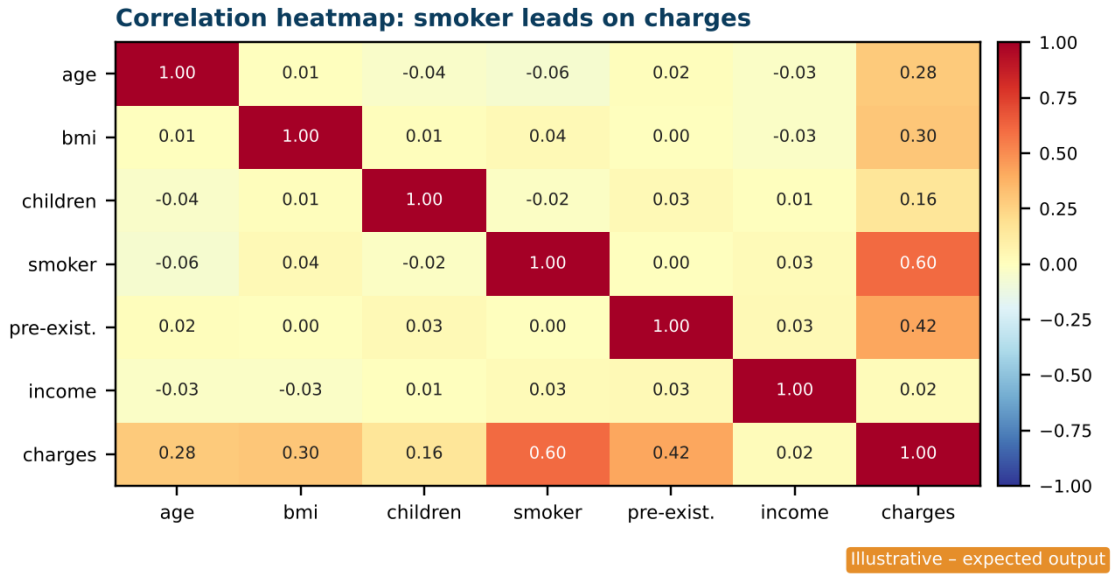
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2500 entries, 0 to 2499
Data columns (total 10 columns):
 #  Column          Non-Null  Dtype
 0  age             2500      int64
 1  sex             2500      object
 2  bmi             2425      float64
 3  children        2500      int64
 4  smoker          2500      object
 5  region          2500      object
 6  city_tier       2500      object
 7  annual_income   2500      int64
 8  pre_existing    2500      object
 9  charges         2500      float64
memory usage: 195.4+ KB
```

Illustrative – expected output

Data Preprocessing & Exploratory Analysis



Illustrative – expected output



Illustrative – expected output

Cleaning

Missing BMI filled with the median; duplicates removed; outliers checked with IQR, high charges kept.

Transformation

sex, smoker, pre-existing coded 0/1; region, city tier one-hot; numbers scaled; log of charges tested.

Feature Selection

Correlation and importance checked; 7 of 9 features kept (sex and region add little).

Module-wise Description

1 Data Preprocessing & EDA

Purpose: Clean, encode and explore the cost dataset

Input → Output: Raw records → clean data and plots

Developed by: S. Trividha

2 Cost Prediction Models

Purpose: Train and compare three regression models

Input → Output: Clean data → best model and metrics

Developed by: K. Deepthi

3 NGO Dataset & Matching Engine

Purpose: Curate aid data and apply eligibility rules

Input → Output: Profile + cost → ranked aid list (₹)

Developed by: M. Swetha

4 Dashboard & Integration

Purpose: Combine prediction, matching and charts

Input → Output: User form → cost, aid and analytics

Developed by: All members

Algorithms & Technologies Used

Linear Regression

Why chosen
Simple, fast baseline that is easy to explain

Key parameters
Scaled numeric inputs; one-hot encoded categories

Random Forest Regression

Why chosen
Handles non-linear effects and interactions; stable

Key parameters
300 trees, max depth 10, min leaf 3

Gradient Boosting Regression

Why chosen
Learns from earlier errors; strong on tabular data

Key parameters
300 trees, learning rate 0.05, depth 3

Selection criteria: lowest MAE and RMSE, highest R^2 on the test set, checked with 5-fold cross-validation. The aid matcher is rule-based on purpose, so every result can be explained.

Implementation / Model Building

Feature Selection

7 inputs: age, BMI, children, smoker, city tier, income, pre-existing condition

Preprocessing Pipeline

Median imputation, 0/1 and one-hot encoding, scaling, all inside one Pipeline

Data Splitting

80% train (2,000 rows) / 20% test (500 rows); 5-fold CV on the train part

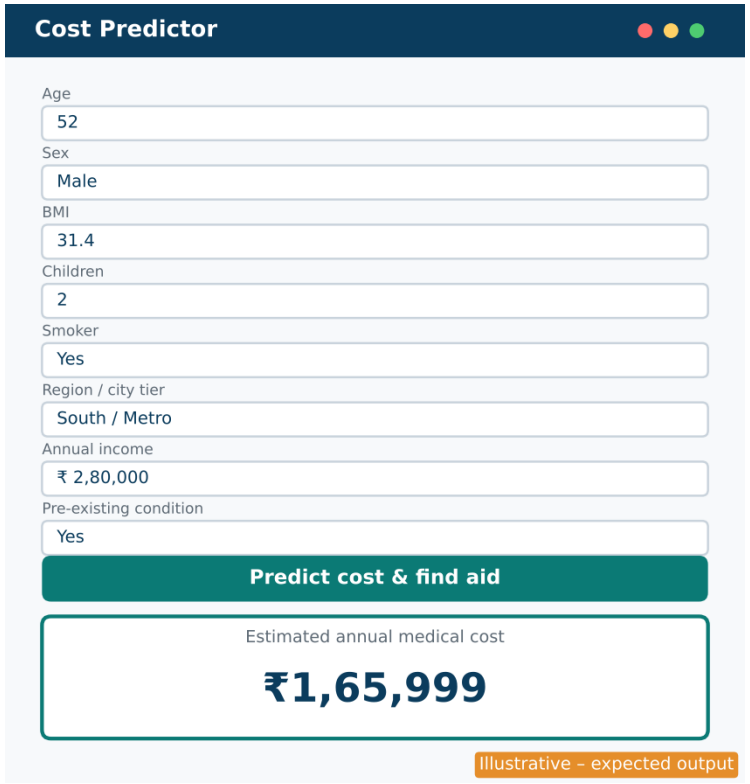
Configuration

random_state=42; grid search over number of trees, depth and learning rate

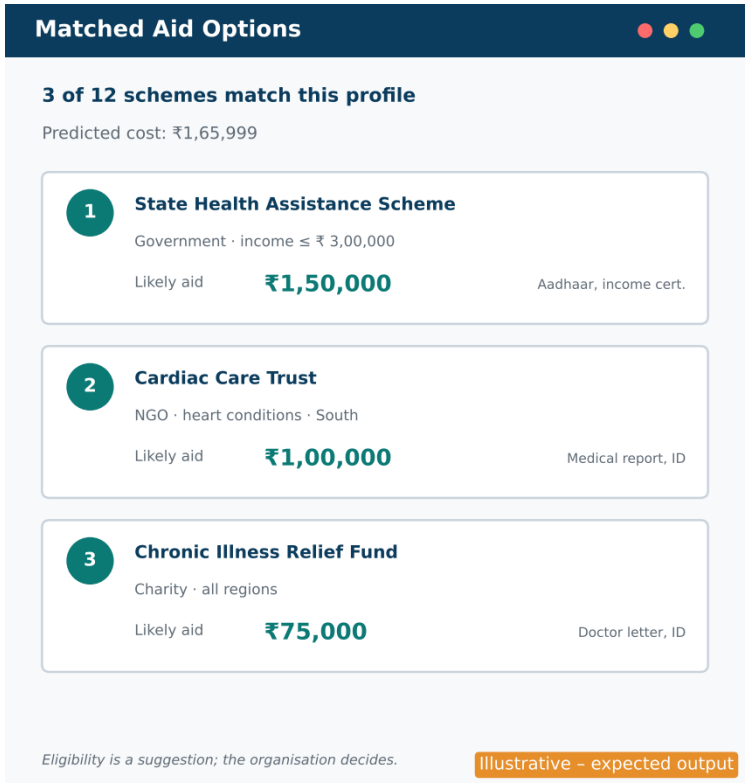
Code snippet / pseudocode

```
# Pipeline: preprocessing + model
pre = ColumnTransformer([
  ("num", StandardScaler(), num_cols),
  ("cat", OneHotEncoder(), cat_cols)])
X_train, X_test, y_train, y_test = \
  train_test_split(X, y, test_size=0.2,
                   random_state=42)
model = Pipeline([("pre", pre),
  ("gbr", GradientBoostingRegressor(
    n_estimators=300, learning_rate=0.05,
    max_depth=3))])
model.fit(X_train, y_train)
```

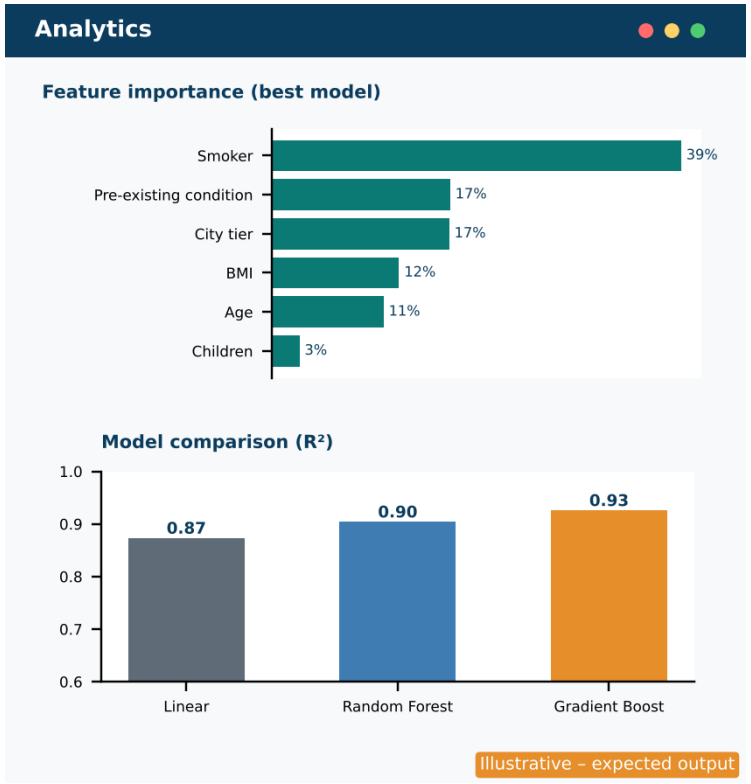
Working Prototype / Output Screens



1. Cost Predictor
Profile form and yearly cost estimate (₹)

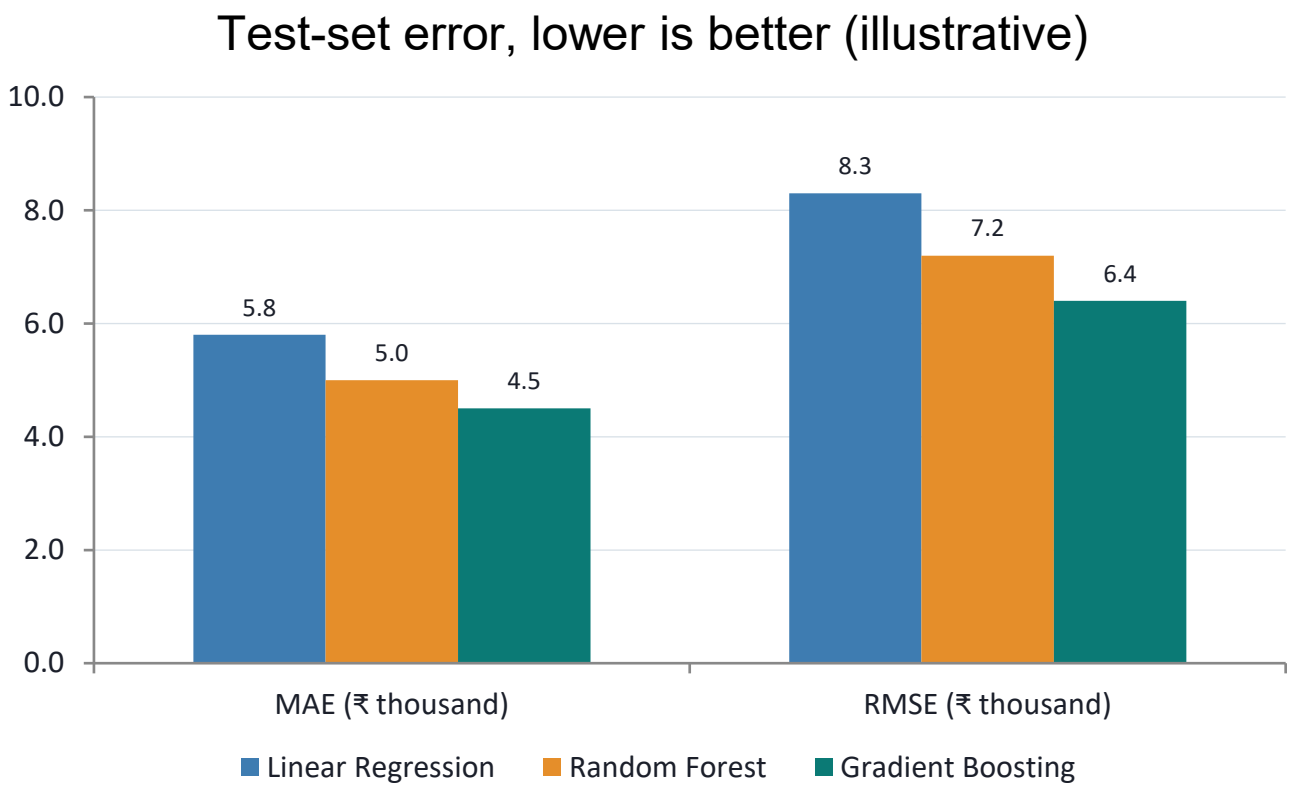


2. Matched Aid Options
Eligible NGOs and schemes with likely aid and documents



3. Analytics
Feature importance and model comparison charts

Results & Performance Comparison

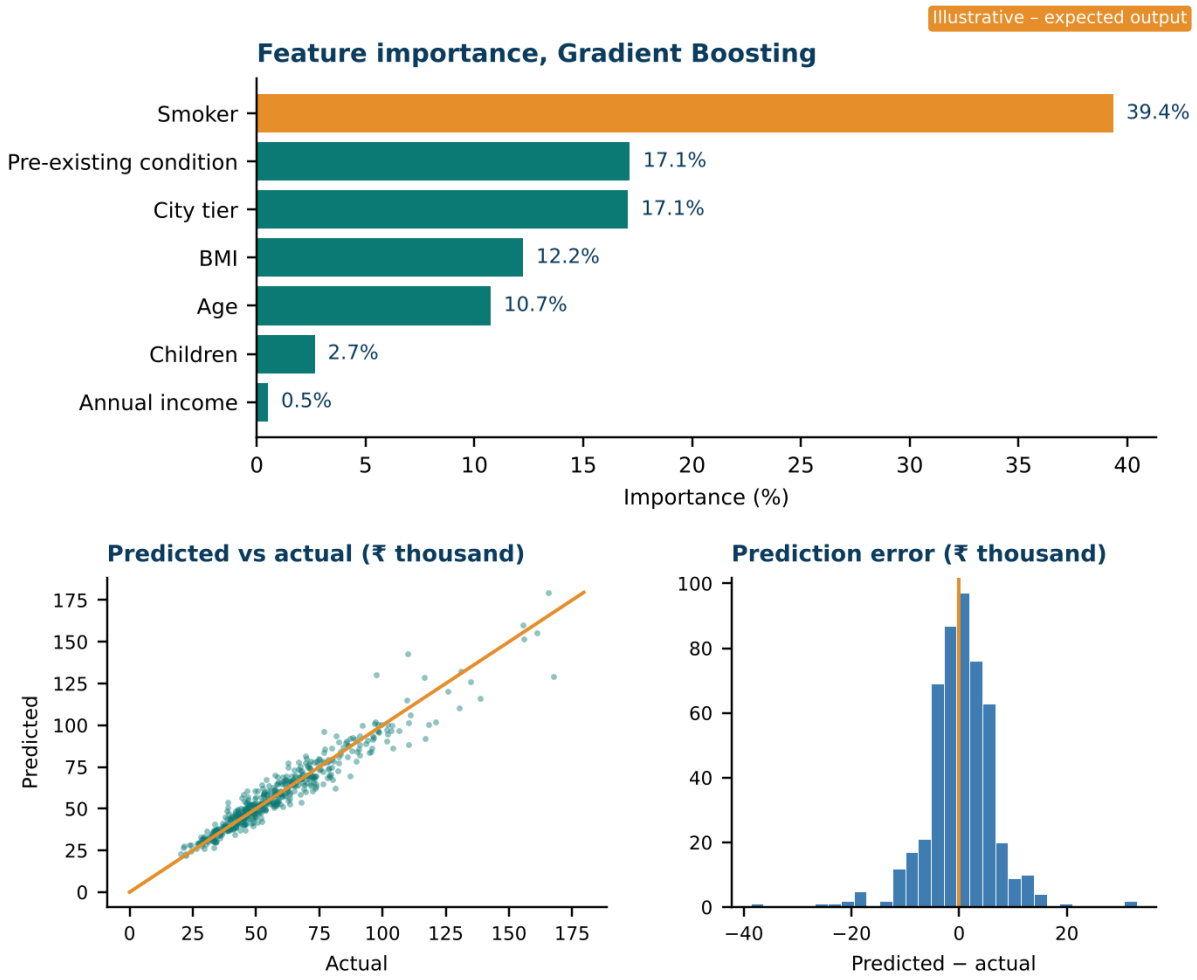


Metric	Linear Reg.	Random Forest	Grad. Boosting
MAE (₹)	₹5,792	₹5,048	₹4,456
RMSE (₹)	₹8,331	₹7,242	₹6,364
R^2	0.87	0.90	0.93
Adj. R^2	0.87	0.90	0.92
5-fold CV R^2	0.88	0.90	0.92

Best: Gradient Boosting – R^2 = 0.93, MAE = ₹4,456

Illustrative values from a simulated Indian-profile dataset. Replace with real results after the project is built.

Analysis & Key Findings



92.6%

of cost variation explained (R^2) by Gradient Boosting

Interpretation: Predictions stay close to actual charges; a typical error is about ₹4,456.
Error analysis: The largest errors are for rare, very high-cost patients with few training examples.
Key drivers: Smoking status (39%), pre-existing condition, city tier, BMI and age.

Testing & Validation

Test ID	Test Description	Expected Result	Actual Result	Status
TC-01	Income equal to scheme limit	Scheme shown (limit inclusive)	Scheme shown	Pass
TC-02	Income above scheme limit	Scheme skipped	Scheme skipped	Pass
TC-03	Scheme with condition “any”, region “all”	Shown for every profile	Shown for all test profiles	Pass
TC-04	Negative age or empty income	Clear error, no crash	Error message shown	Pass

Functional Testing

Unit tests for each rule; end-to-end test from form to result

Validation Method

20% hold-out set and 5-fold cross-validation; matcher checked against hand-worked answers

Performance Testing

Prediction and matching answer in under a second on a laptop

Challenges & Limitations

Challenges Faced

- Skewed, very high charges → kept outliers and tested a log transform
- Different rule formats across NGOs → one standard rule format with source and date
- Missing BMI values → median imputation inside the pipeline
- Little public Indian cost data → built an India-adapted dataset in ₹

Limitations

- Dataset is self-prepared, so costs are a rough guide and not a bill
- Only about 2,500 records, so models can overfit
- NGO data is collected by hand and needs regular updates
- Eligibility is a suggestion; the organisation decides

Conclusion

1. Built a cost predictor and an aid matcher on an India-adapted medical cost dataset (₹), shown in one dashboard.

2. Gradient Boosting was best: R^2 0.93 and MAE ₹4,456, ahead of Random Forest (0.90) and Linear Regression (0.87).

3. All four objectives met: cleaning and EDA, model comparison, NGO dataset with rule-based matching, and the dashboard.



Final Outcome

0.93

R^2 of the best model (Gradient Boosting)
Objectives achieved: 4 of 4

Future Scope & Applications

Improve Performance
Tune further, try XGBoost, add SHAP explanations

Expand Data / Features
Real Indian claims data, more illnesses and states

Deploy & Build an Interface
Public web app with Hindi and Telugu, plus a mobile version

Scale & Secure
Cloud hosting, no storage of personal data, regular scheme updates

Applications

- Hospital help desks and patient counsellors
- NGOs and trusts screening applicants
- Government and insurance health portals
- Families planning medical expenses

Team Contribution

Reg. No.	Name	Contribution / Modules Handled
2420030010	S. Trividha	Data collection, preprocessing and EDA (Module 1); report writing
2420030484	K. Deepthi	Model building and evaluation (Module 2); results and analysis
2420030116	M. Swetha	NGO and welfare dataset, rule-based matching engine (Module 3)

Dashboard and integration (Module 4) are built together by all members.

References

[1] Orji et al., “Machine learning for an explainable cost prediction of medical insurance,” arXiv, 2024.

[2] “Medical insurance cost prediction using machine learning,” Int. J. Res. Appl. Sci. Eng. Technol. (IJRASET), 2023.

[3] Hong et al., “Predicting a need for financial assistance in emergency department care,” PMC/NCBI, 2025.

[4] M. Rajpoot, A. Thakur, and S. Kumar, “Government schemes AI recommendation system,” 2025.

[5] U. S. Sabarinath and A. Mathew, “Medical insurance cost prediction,” IJDCN, vol. 4, no. 4, pp. 1–4, 2024.

[6] D. S. AbdElminaam et al., “An efficient framework for predict medical insurance costs using machine learning,” J. Comput. Commun., vol. 3, no. 2, pp. 55–64, 2024.

[7] P. Mohanraj et al., “AI enabled system for simplified access to government schemes for all citizens,” IJEETR, vol. 8, no. 2, pp. 4365–4375, 2026.



Project Links

GitHub: github.com/SuryadevaraTrividha/-DS_S7_T02_Medical-Insurance-Cost-Predictor
Dataset: Self-prepared, kept in the project repository
Demo video: To be added

THANK YOU

Questions & Discussion

S. Trividha, K. Deepthi, M. Swetha

Guide: Ms. B. Pratyusha, Assistant Professor,
Dept. of CSE, KLH CSE Bowrampet Campus